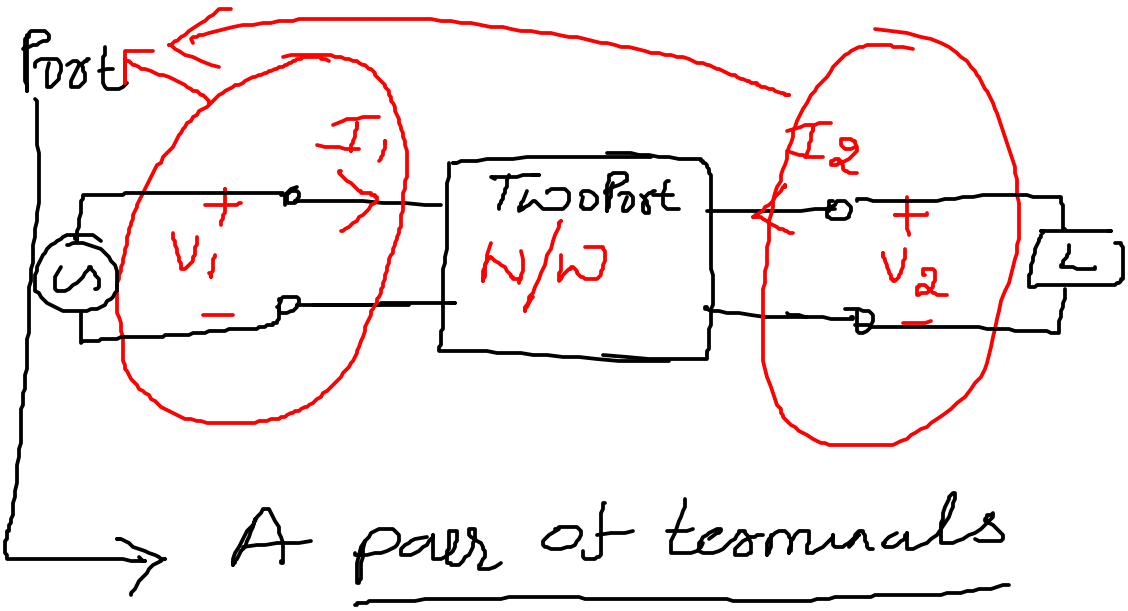
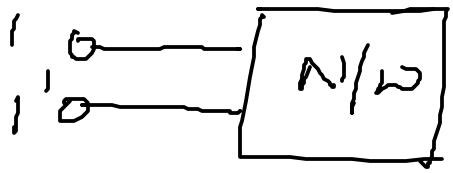
UNIT-5 → Two Port Network

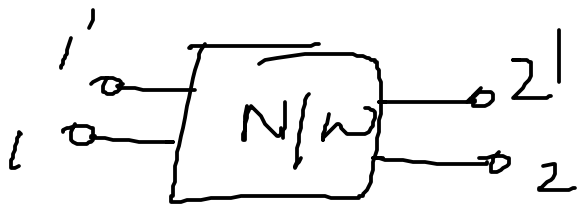


→ A pair of terminals

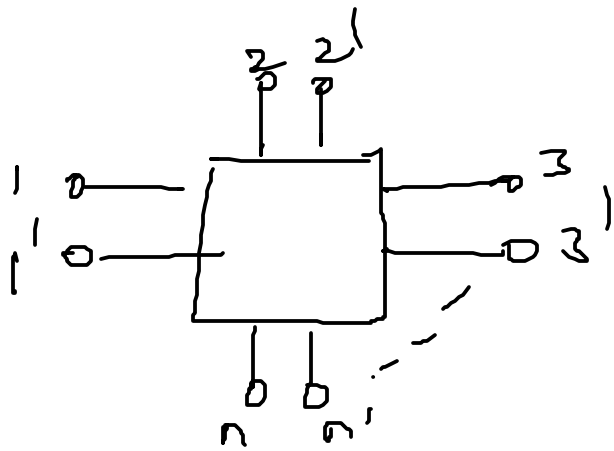
N/W can be



→ one port n/w



→ two port n/w



→ multi port n/w

Standard Polarities in two port n/w

Current must enter

For two port n/w

6 Parameters → (1) Impedance Parameter (Z)

(2) Admittance Parameter (Y)

(3) Hybrid Parameter (H)

(4) Transmission Parameter / ABCD parameter.

(5) → Inverse Hybrid parameter

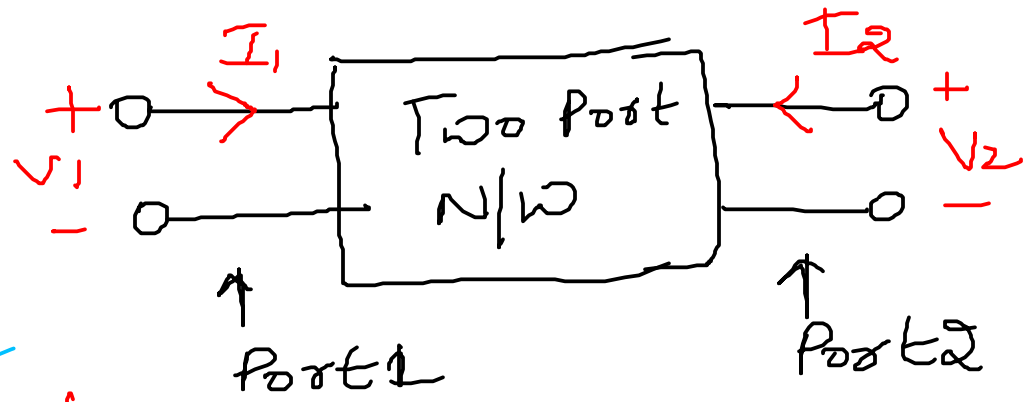
(6) → Inverse Transmission parameter

S-Parameter → for high frequency operations

① Z parameter / Impedance parameter

② open ckt parameter.

Std two port n/w.



3 forms

Functional form → $V_1 = f(I_1, I_2)$

$V_2 = f(I_1, I_2)$

express Voltage in terms of current

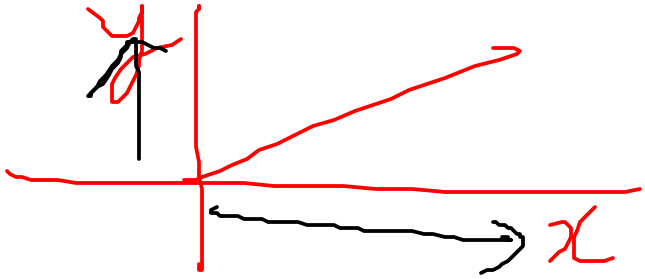
↳ Z Parameter

$$V = IR = IZ$$

$$V = IZ$$

Independent variable → I

dependent variable → V.



$y = m x$ ← slope

dependent variable

independent variable

(2) Equational form

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \longrightarrow (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \longrightarrow (2)$$

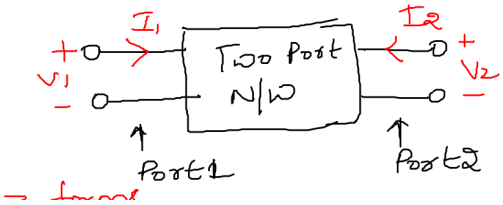
(3) Matrix form

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$$

$$[V] = [Z][I]$$

Calculate Individual Z-parameter

Std two port n/w



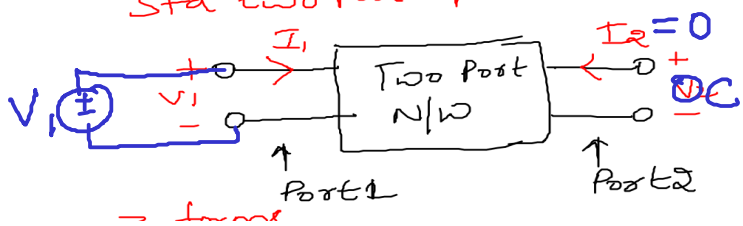
$$V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (2)$$

Make $I_2 = 0 \rightarrow$ make Port 2 $\rightarrow$ OC $\Rightarrow I_2 = 0$

Port 1 $\rightarrow$ connect V/g Src

Std two port n/w



$I_2 = 0 \Rightarrow$ eq^n (1) $\Rightarrow V_1 = Z_{11} I_1 \Rightarrow \boxed{Z_{11} = \frac{V_1}{I_1}}$

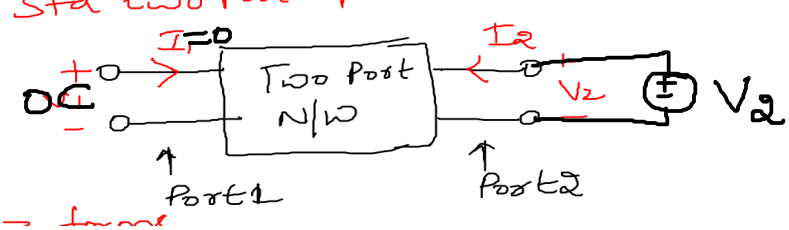
open ckt input impedance

eq^n (2) $\Rightarrow V_2 = Z_{21} I_1 \Rightarrow \boxed{Z_{21} = \frac{V_2}{I_1}}$

open ckt Transfer impedance

Make $I_1 = 0 \Rightarrow$ open ckt Port 1

Std two port n/w



$I_1 = 0 \Rightarrow$ eqn (1) $\Rightarrow V_1 = Z_{12} I_2$

$$\Rightarrow \boxed{Z_{12} = \frac{V_1}{I_2}}$$

open ckt transfer impedance

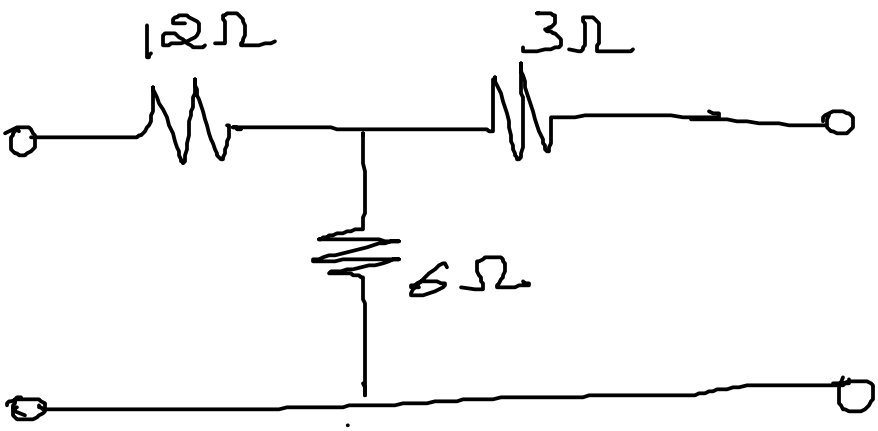
$\rightarrow$ eqn (2) $\Rightarrow V_2 = Z_{22} I_2$

$$\boxed{Z_{22} = \frac{V_2}{I_2}}$$

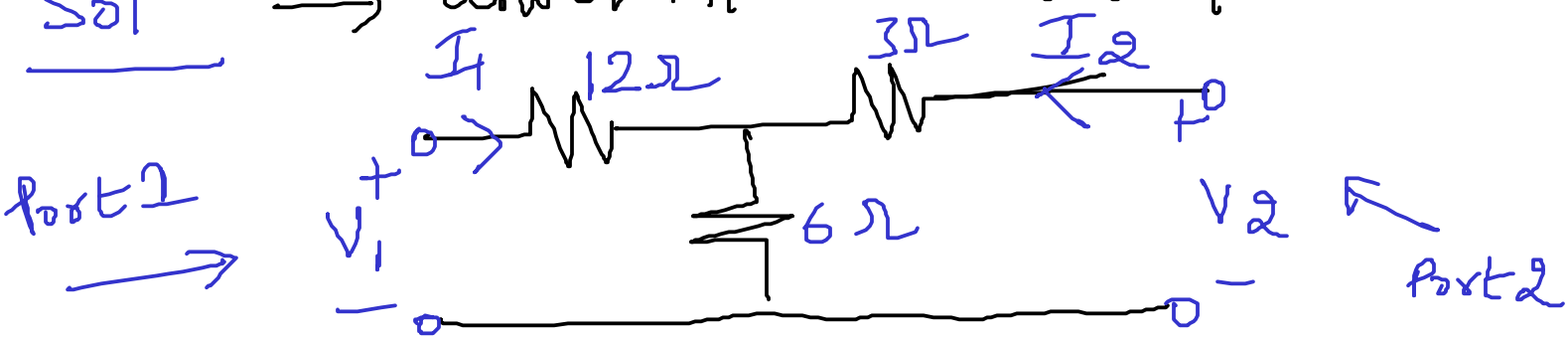
open ckt output impedance

Ex 1:- For the given n/w find Z-parameters

Open ckt parameter



Sol^n $\rightarrow$ Convert n/w to std n/w



WKT Z parameter is given by

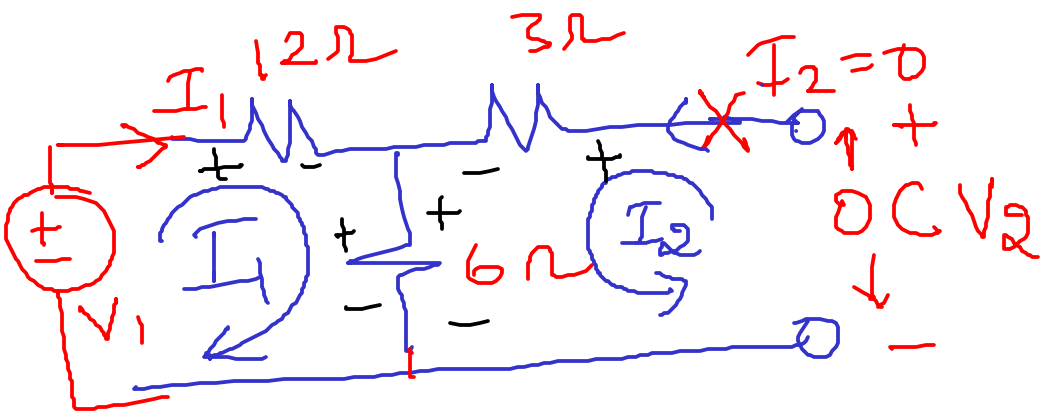
$$V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (2)$$

Make independent variable zero to get Z-parameter.

Here $I_2 = 0 \Rightarrow$ Port 1 $\rightarrow$ Connect a voltage

Port 2 $\rightarrow$ open ckt



↳ Apply KVL for each Mesh & make $I_2 = 0$

KVL at Mesh 1 $\rightarrow$ $-V_1 + I_1 \times 12 + (I_1 + I_2) \times 6 = 0$

make $I_2 = 0 \Rightarrow -V_1 + I_1 \times 12 + I_1 \times 6 = 0$

$V_1 = 18 I_1$

$$Z_{11} \longleftarrow \boxed{\frac{V_1}{I_1} = 18\,\Omega}$$

KVL @ Mesh 2 $\Rightarrow$ $-V_2 + I_2 \times 3 + (I_2 + I_1) \times 6 = 0$

$I_2 = 0 \Rightarrow -V_2 + 6 I_1 = 0$

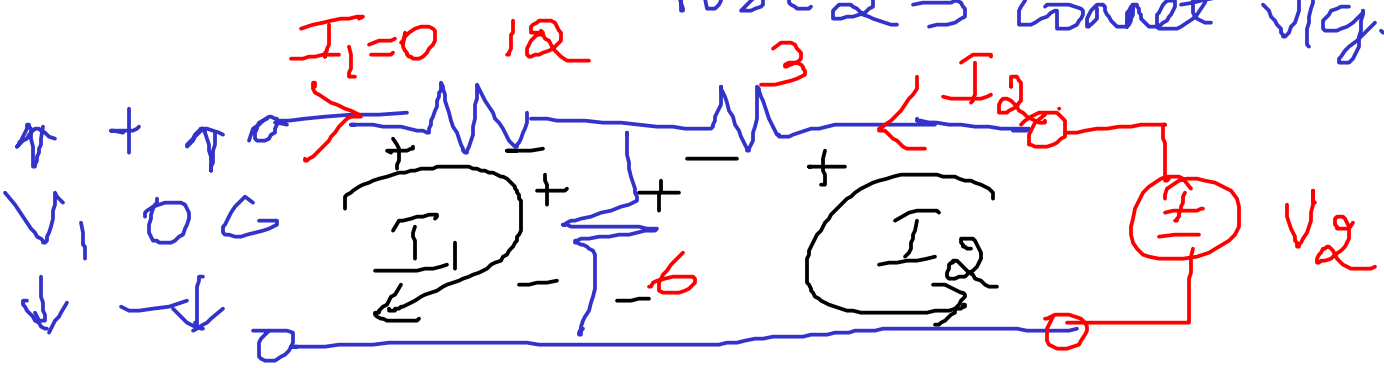
$V_2 = 6 I_1$

$$Z_{21} \longrightarrow \boxed{\frac{V_2}{I_1} = 6\,\Omega}$$

$\frac{V}{I}$ from in Z parameters

make $I_1 = 0 \Rightarrow$ Port 1 $\rightarrow$ OC

Port 2 $\rightarrow$ Connect V/g Soc



Apply KVL @ Mesh 1 & then make $I_1 = 0$

$$-V_1 + 12I_1 + (I_1 + I_2) \times 6 = 0$$

$$I_1 = 0 \Rightarrow -V_1 + I_2 \times 6 = 0 \Rightarrow V_1 = 6I_2$$

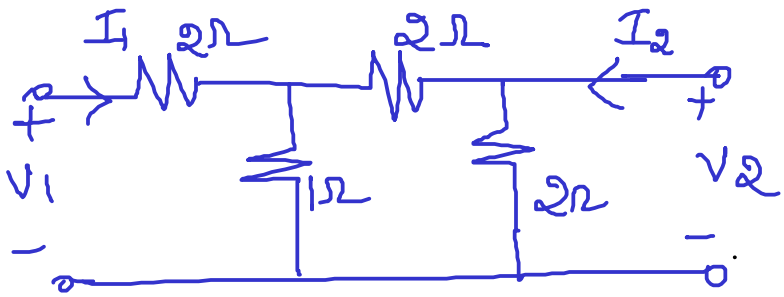
$$\Rightarrow \frac{V_1}{I_2} = 6\Omega$$

$$\Rightarrow \boxed{Z_{12} = 6\Omega}$$

Apply KVL @ Mesh 2 & make $I_1 = 0$

$$\boxed{Z_{22} = 9\Omega}$$

Prob 2:- Obtain Z-parameters for given n/w



Soln :- Std polarity is given

WKT

Z-parameter ⇒

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (1)$$
$$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (2)$$

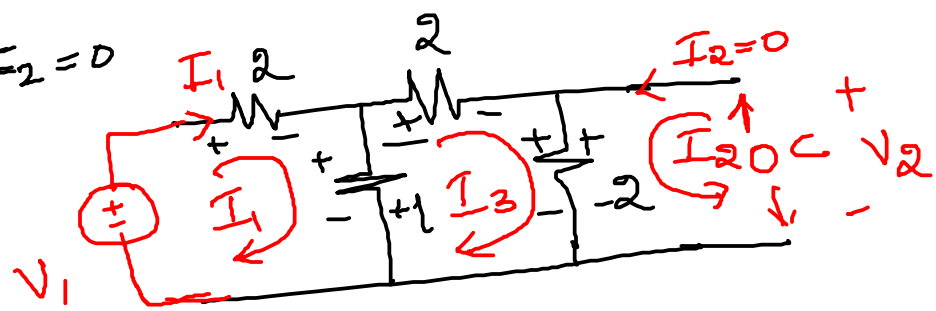
To find Z-parameter make independent variables = 0

V_1, I_1

$V_2 \; I_1$

$V_1, I_1 \; V_2$

Make $I_2 = 0$



KVL @ Mesh 1 → $-V_1 + I_1 \times 2 + (I_1 - I_3) \times 1 = 0$

$I_2 = 0 \Rightarrow$ $V_1 = 3I_1 - I_3 \rightarrow (3)$ → eliminate

KVL @ Mesh 2 → $1 \times (I_3 - I_1) + I_3 \times 2 + 2(I_3 + I_2) = 0$

$I_2 = 0 \Rightarrow$ $5I_3 - I_1 = 0 \rightarrow (4)$ → eliminate

KVL @ Mesh 3 ⇒ $-V_2 + (I_2 + I_3) \times 2 = 0$

$I_2 = 0 \Rightarrow$ $V_2 = I_3 \times 2 \rightarrow (5)$ → eliminates

$V_1, I_1, V_2 \rightarrow Z_{11} \& Z_{21}$ $\xrightarrow{\text{from}}$ Eqn 4 $\rightarrow$ $I_3 = \frac{I_1}{5} \rightarrow$ (6)

eqn (3) $\rightarrow$ $V_1 = 3I_1 - I_3$

Substitute (6) in (3) $\Rightarrow$ $V_1 = 3I_1 - \left(\frac{I_1}{5}\right)$

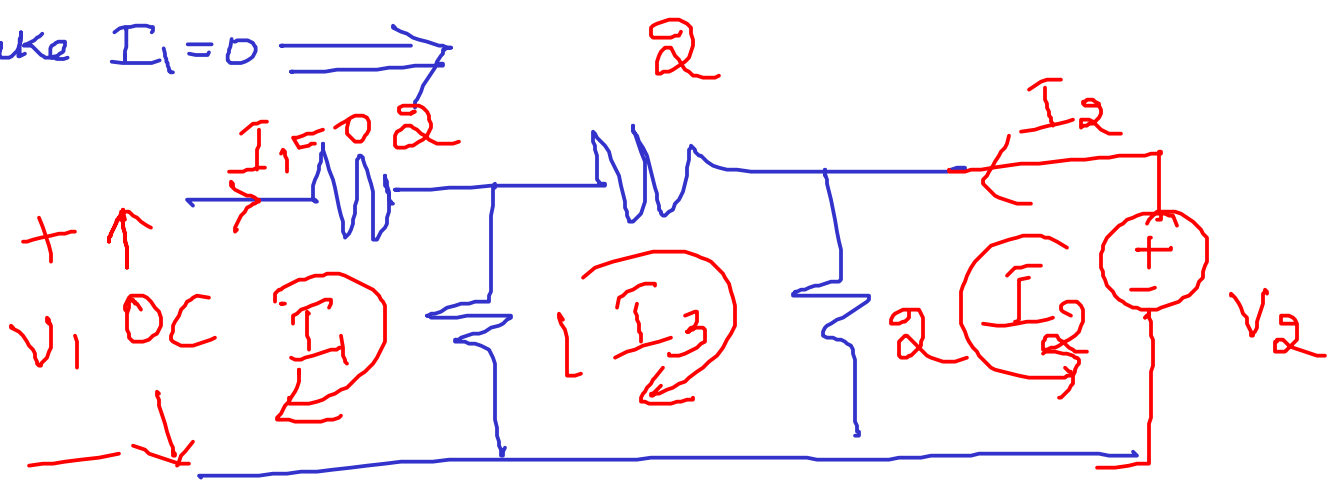
$$Z_{11} \longrightarrow \frac{V_1}{I_1} = \frac{14}{5}\,\Omega$$

Substitute (6) in (4) $\Rightarrow$ $V_2 = I_3 \times 2$

$$V_2 = \frac{I_1}{5} \times 2$$

$$\boxed{Z_{21} = \frac{V_2}{I_1} = \frac{2}{5}\,\Omega}$$

|||ly make $I_1 = 0 \Longrightarrow$



Ans:- $\boxed{Z_{12} = \frac{2}{5}}$, $\boxed{Z_{22} = \frac{6}{5}}$

Y-Parameter :- (Admittance parameter & Shortckt parameter)

$$V = IZ \Rightarrow I = \frac{V}{Z} \Rightarrow I = YV$$

$Y \rightarrow$ Admittance

$\rightarrow$ Y parameter

Express currents in terms of v/g

(1)

Admittance parameter in functional form

$$I = V$$

$$\Rightarrow \text{Port 1} \rightarrow I_1 = f(V_1, V_2)$$

$$\text{Port 2} \rightarrow I_2 = f(V_1, V_2)$$

② In equational form

$$I_1 = y_{11} V_1 + y_{12} V_2$$
$$I_2 = y_{21} V_1 + y_{22} V_2$$

③ Matrix form

$$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

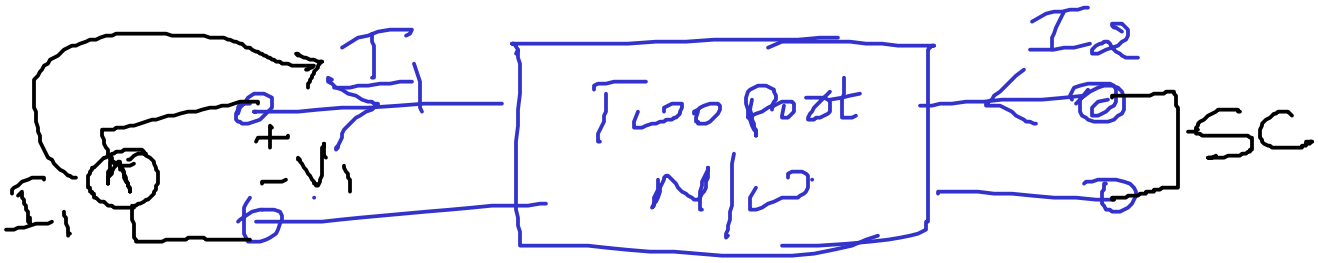
To calculate $\boxed{Y \text{ parameter}}$

$$** \begin{cases} I_1 = y_{11} V_1 + y_{12} V_2 \rightarrow ① \\ I_2 = y_{21} V_1 + y_{22} V_2 \rightarrow ② \end{cases}$$

Make independent variable = 0

$V_2 = 0 \longrightarrow$ Port 1 $\longrightarrow$

Port 2 $\longrightarrow$ Short ckt



$V_2 = 0 \Rightarrow$ (1) $\Rightarrow I_1 = y_{11} V_1$

$$\Rightarrow \frac{I_1}{V_1} = y_{11} \quad \mho$$

Short ckt input Admittance

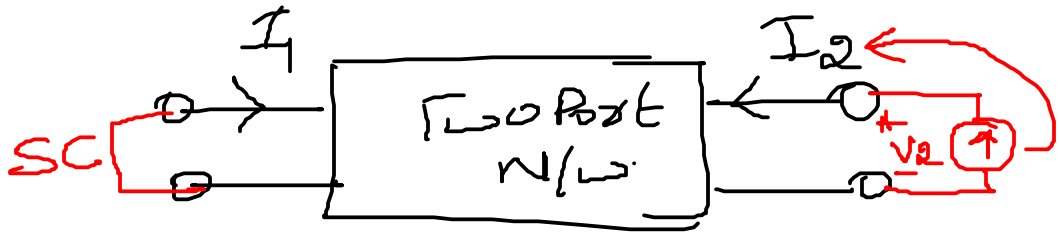
$V_2 = 0 \Rightarrow$ (2) $\Rightarrow I_2 = y_{21} V_1$

$$\Rightarrow y_{21} = \frac{I_2}{V_1} \quad \mho \text{ (S)}$$

Short ckt Transfer Admittance

Make $V_1 = 0 \Rightarrow$ Port 1 $\rightarrow$ Short ckt

Port 2 $\rightarrow$ Connect Current Source.



$V_1 = 0 \Rightarrow (1) \Rightarrow I_1 = Y_{12} V_2$

$$y_{12} = \frac{I_1}{V_2} \ \mho \text{ (or) } S$$

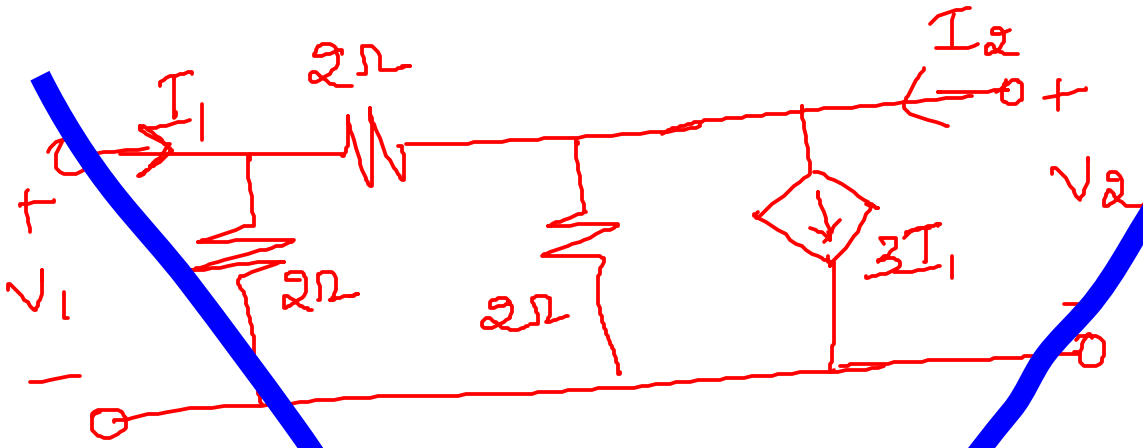
Short ckt Transfer Admittance

$V_1 = 0 \Rightarrow (2) \Rightarrow I_2 = y_{22} V_2$

$$\Rightarrow y_{22} = \frac{I_2}{V_2} \ \mho \text{ (or) } S.$$

$\boxed{\text{Short ckt output Admittance}}$

EX :- Find Y-parameters for the given n/w



Sol :- To cal Y parameters

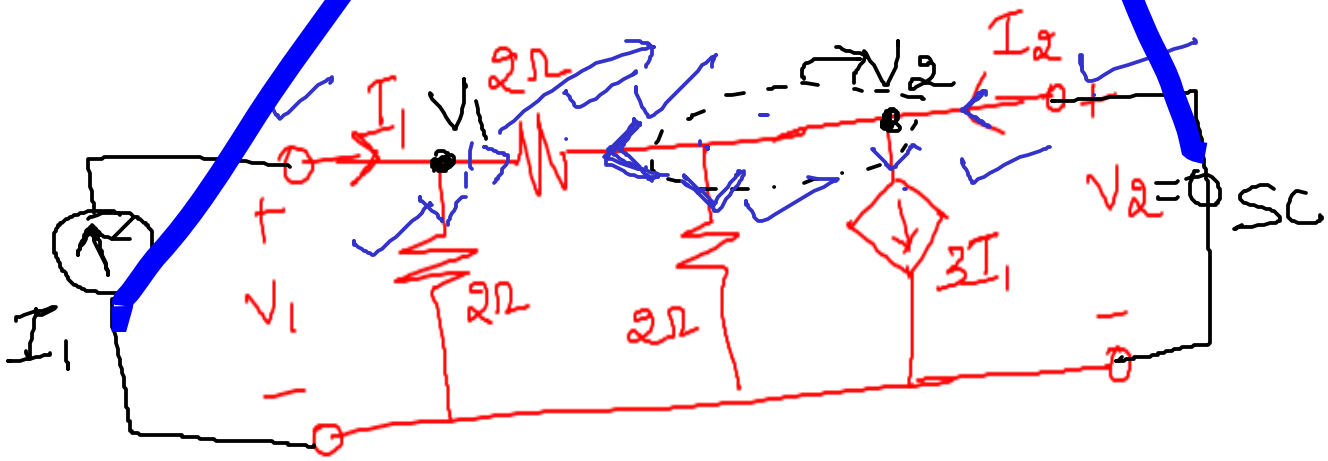
WKT $\rightarrow$ $I_1 = y_{11}V_1 + y_{12}V_2 \rightarrow (1)$

$I_2 = y_{21}V_1 + y_{22}V_2 \rightarrow (2)$

$\rightarrow$ Make independent variables $= 0$

$V_2 = 0 \rightarrow$ Port 2 $\rightarrow$ Short ckt

Port 1 $\rightarrow$ Const current source



Apply KCL for each principal node &

make $V_2 = 0$

KCL @ node V_1

$$\rightarrow -I_1 + \frac{V_1}{2} + \frac{V_1 - V_2}{2} = 0$$

Put $V_2 = 0 \Rightarrow -I_1 = -V_1$

$$\boxed{\frac{I_1}{V_1} = 1\,\mho}$$

$\boxed{y_{11}}$

$I_1 = V_1$

$\rightarrow$ (*)

KCL @ node V_2

$$\frac{V_2 - V_1}{2} + \frac{V_2}{2} + 3I_1 - I_2 = 0$$

make $V_2 = 0 \Rightarrow \dfrac{-V_1}{2} + 3I_1 - I_2 = 0$

$\rightarrow$ (4)

V_1 I_1 & I_2

Substitute (3) in (4) $\Rightarrow$ $\boxed{\dfrac{I_2}{V_1} = \dfrac{5}{2} \mho = y_{21}}$

Case 2 $V_1 = 0$

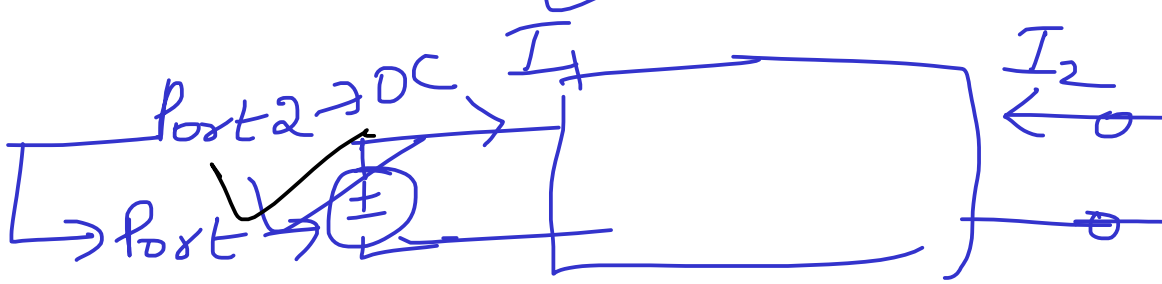
Ans $\longrightarrow$ $\boxed{y_{12} = \dfrac{-1}{2} \mho \ \& \ y_{22} = \dfrac{-1}{2}}$

Z-Parameter $\longrightarrow$ $V = Z I$

$\longrightarrow$ $V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

$I_2 = 0$



$I_1 = 0$

$\longrightarrow$ Apply KVL

Y - Parameter $\Rightarrow$ $I = VY$

$$I_1 = y_{11} V_1 + y_{12} V_2$$

$$I_2 = y_{21} V_1 + y_{22} V_2$$

$V_1 = 0 \rightarrow$ Port 2 $\rightarrow$ SC

$\rightarrow$ Port 1 $\rightarrow$ current SOC

$V_2 = 0$

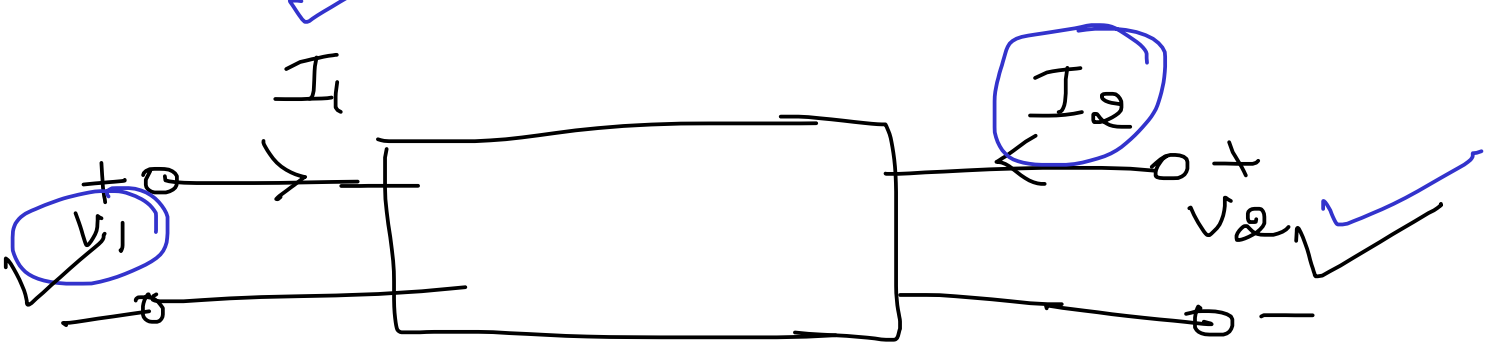
KCL $\longrightarrow$ Solve

Hybrid parameter

functional form

$$V_1 = f(I_1, V_2)$$

$$I_2 = f(I_1, V_2)$$



Z - Parameter $V = IZ$

Equation form

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad (2)$$

Matrix form

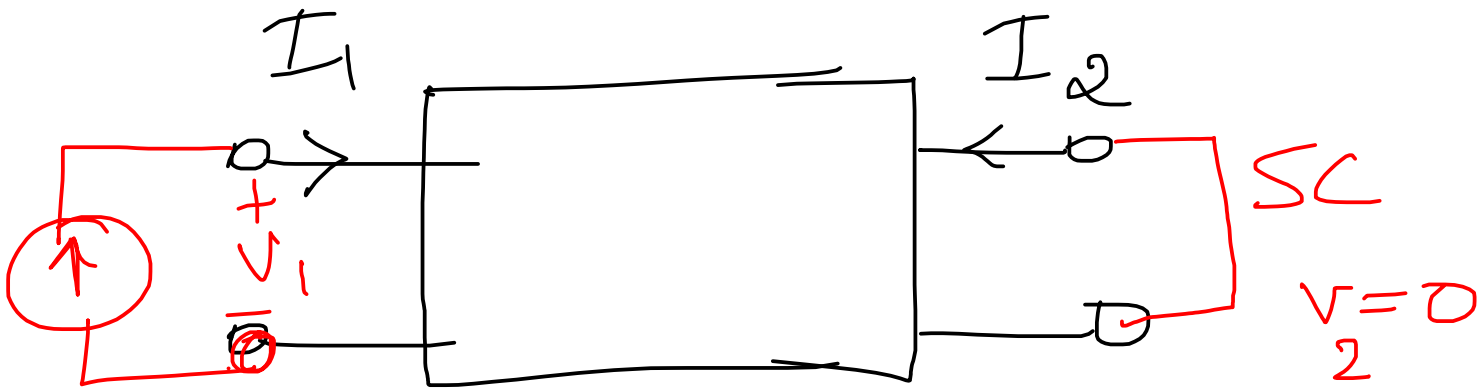
$$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ V_2 \end{pmatrix}$$

To get h parameter evaluate

independent variable to zero

$V_2 = 0 \rightarrow$ Port 1 $\rightarrow$

$\rightarrow$ Port 2 $\rightarrow$ SC



$V_2 = 0 \Rightarrow (1) \Rightarrow V_1 = h_{11} I_1$

$$h_{11} = \frac{V_1}{I_1} \Bigg|_{V_2 = 0} \quad \Omega$$

$\rightarrow Z_{11}$

Short ckt i/p Impedence

eq^n (2) $\rightarrow$ $I_2 = h_{21} I_1$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} \quad \text{(no unit)}$$

Short ckt f/d current gain

Port 1 → OC

Illy $I_1 = 0$ — Port 2

① $\Rightarrow$ $V_1 = h_{12} V_2$

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0} \quad \text{nounit}$$

open ckt rev V/y gain

② $\Rightarrow$ $I_2 = h_{22} V_2$

$$h_{22} = \frac{I_2}{V_2}\Big|_{I_1=0}$$

$v(s)$

o/p

open ckt admittance

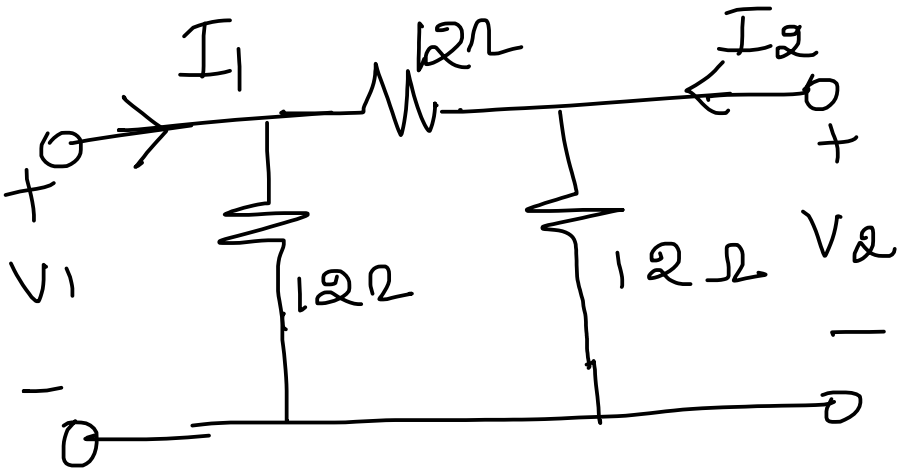
z_{11} $y_{11} \rightarrow$

z_{12} $y_{12} \rightarrow$

z_{21} $y_{21} \rightarrow$

z_{22} $y_{22} \rightarrow$

Determine h parameter for given 2 port n/w



Sol^n:- → h-parameter

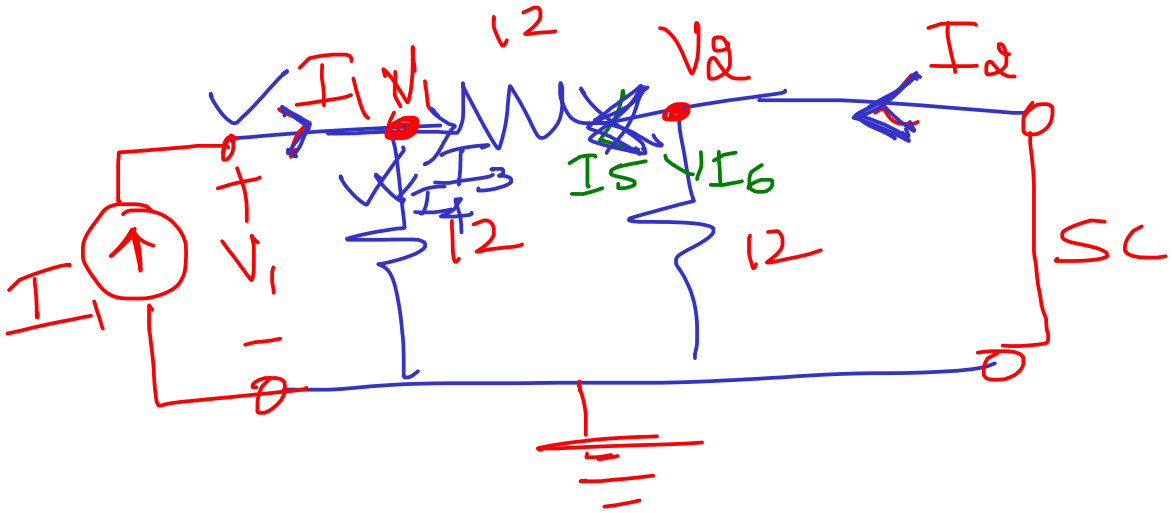
$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

V_1, I_2

$\frac{I_2}{I_1}$

make $V_2 = 0$ → Port 2 → SC
Port 1 → Current Sc



KCL @ $V_1 \Rightarrow (V_1 >> V_2)$

$$-I_1 + \frac{V_1}{12} + \frac{V_1 - V_2}{12} = 0$$

$$V_2 = 0 \Rightarrow -I_1 + \frac{V_1}{6} = 0$$

$$\boxed{h_{11} = \frac{V_1}{I_1} = 6\,\Omega}$$ // → 3

KCL @ $V_2 \Rightarrow (V_2 >> V_1)$

$$-I_2 + I_5 + I_6 = 0$$

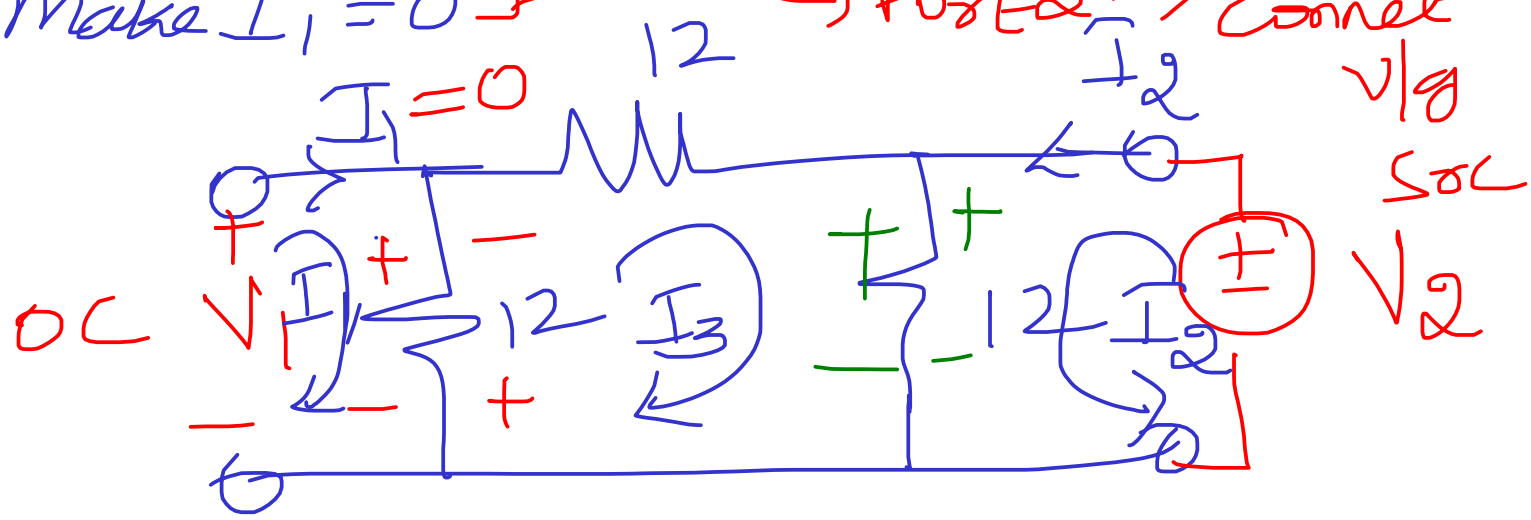
$$-I_2 + \frac{V_2 - V_1}{12} + \frac{V_2}{12} = 0$$

$$V_2 = 0 \Rightarrow -I_2 - \frac{V_1}{12} = 0 \rightarrow 4$$

use eqn 3 in eqn 4 $\Rightarrow -I_2 - \frac{(I_1 \times 6)}{12} = 0$

$$h_{21} = \frac{I_2}{I_1} = \frac{-1}{2}$$

Make $I_1 = 0$ → Port 1 → OC
→ Port 2 → connect V/g SOC



KVL @ Mesh 1

$$-V_1 + (I_1 - I_3) * 12 = 0$$

$$I_1 = 0 \Rightarrow -V_1 - 12 I_3 = 0$$

$$-V_1 - 12 I_3 = 0$$

KVL @ Mesh 2

$$+12(I_3 - I_1) + 12 I_3 + 12 \times (I_3 + I_2) = 0$$

$I_1 = 0 \Rightarrow$ $36 I_3 + 12 I_2 = 0$

$3 I_3 + I_2 = 0$

$\rightarrow 6$

KVL @ mesh 3

$-V_2 + 12(I_2 + I_3)$

$V_2 = 12 I_2 + I_3$

$\rightarrow 7$

$V_1 = h_{11} I_1 + h_{12} V_2$

$I_2 = h_{21} I_1 + h_{22} V_2$

$V_2 \quad V_1 \quad I_2$

$V_1 \quad V_2$

$I_2 \quad V_2$

$h_{22} = \frac{I_2}{V_2}$

$$6 \Rightarrow I_3 = -\frac{1}{3} I_2 \rightarrow 8$$

Substitute 8 in 7

$$\frac{I_2}{V_2} = \frac{1}{8} \mho$$

$$\cancel{\frac{1}{22}}$$

$$5 \Rightarrow I_3 = -\frac{1}{12} V_1 \rightarrow 9$$

Substitute (9) in (7) & Solve.

$$-V_1 - 12 I_3 = 0 \rightarrow (4)$$

I_2

$$I_3 = \frac{1}{\quad}$$

$$V_2 = 12 I_2 + 12 I_3$$

$$I_3 = \frac{-1}{3} I_2$$

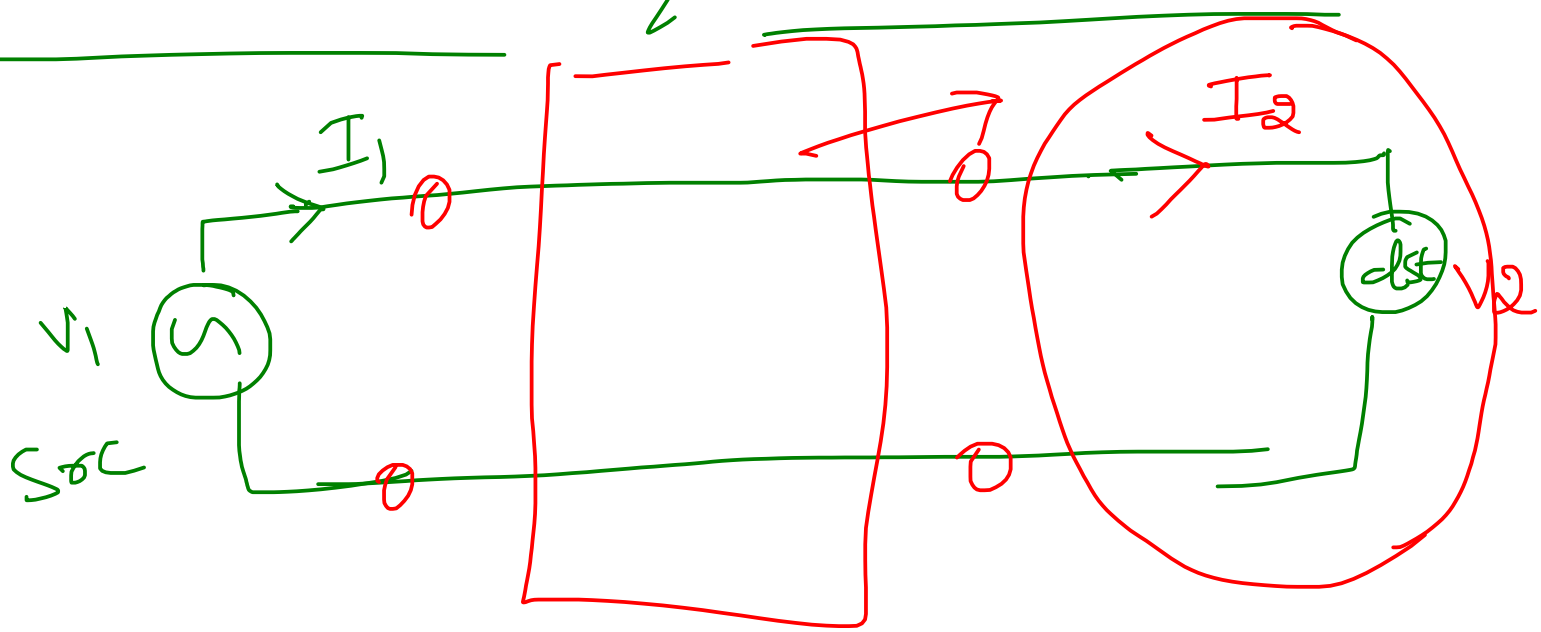
$V_1 \quad V_2$

$$I_2 = \frac{1}{8}$$

V_2

$$h_{12} = \frac{1}{2}$$

Transmission Parameter / ABCD Parameter



~~I change~~

functional form

$$V_1 = f(V_2, -I_2)$$
$$I_1 = f(V_2, -I_2)$$

$-I_2$ → I change

Equation form

$$V_1 = A V_2 + B(-I_2) \rightarrow (1)$$

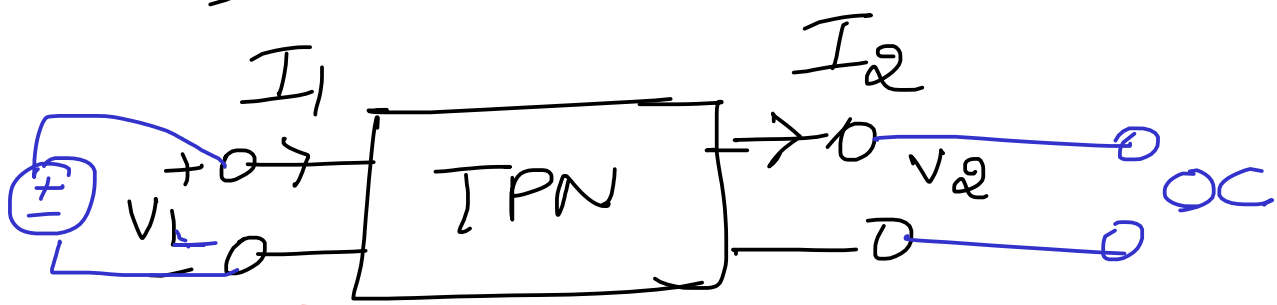
$$I_1 = C V_2 + D(-I_2) \rightarrow (2)$$

Matrix form

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}$$

To find ABCD parameters $\Rightarrow$ make independent variable

$I_2 = 0 \Rightarrow$ Port 2 $\rightarrow$ OC
Port 1 $\rightarrow$



$I_2 = 0 \Rightarrow V_1 = A V_2$

$$\Rightarrow A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$

$\frac{V}{I}$	$\frac{I}{V}$	$\frac{I}{I}$	$\frac{V}{V}$

Rev

open ckt / v|g gain

v|g ratio

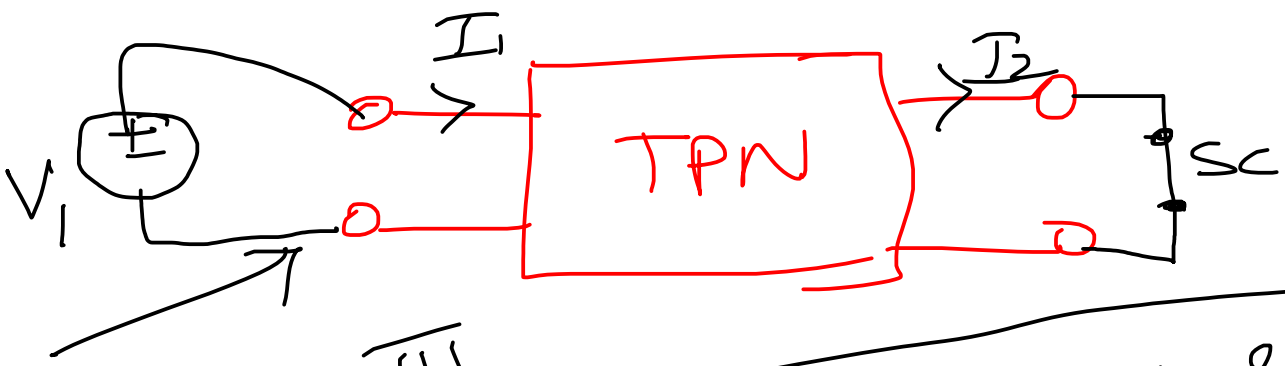
eq^n ② $\Rightarrow$ $I_1 = C V_2$

$$\Rightarrow C = \left.\frac{I_1}{V_2}\right|_{I_2 = 0}$$

open ckt Transfer admittance

Make $V_2 = 0$ $\longrightarrow$ Port 1

Port $\longrightarrow$ SC



change III

connect v|g src for each case

$V_2 = 0 \Rightarrow$ ① $\Rightarrow$ $V_1 = -B I_2$

$$B = \left.\frac{V_1}{(-I_2)}\right|_{V_2=0}$$

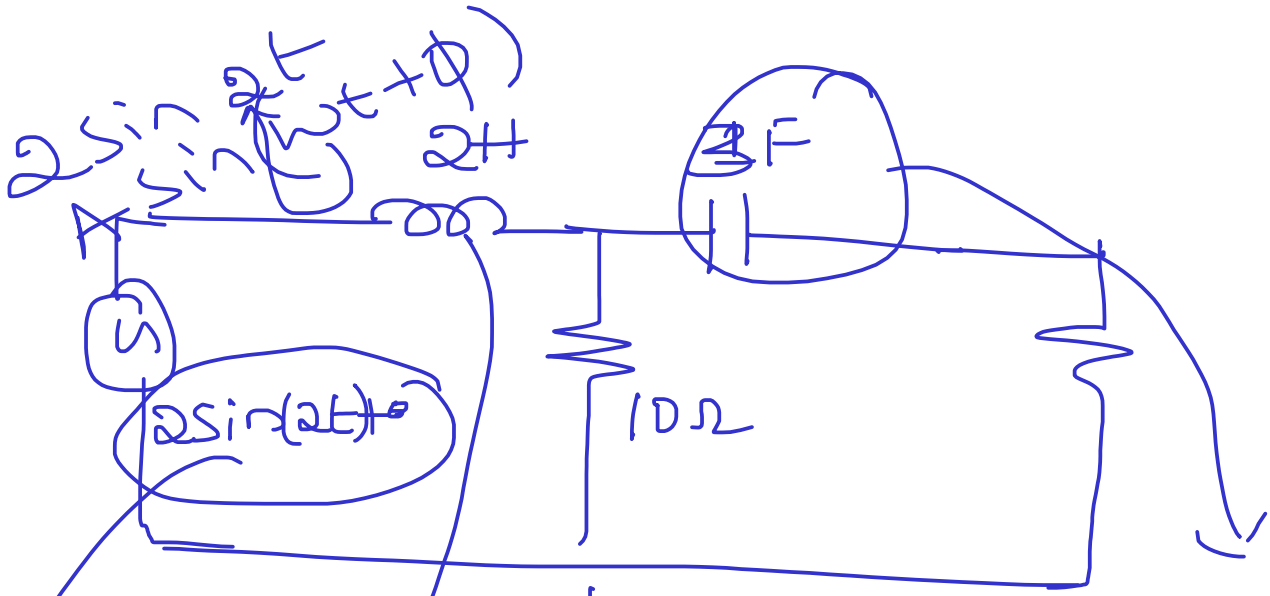
negative Short ckt Transfer Impedance

$V_2 = 0 \Rightarrow (2) \Rightarrow I_1 = -DI_2$

$$D = \left.\frac{I_1}{(-I_2)}\right|_{V_2=0}$$

negative Short ckt current gain

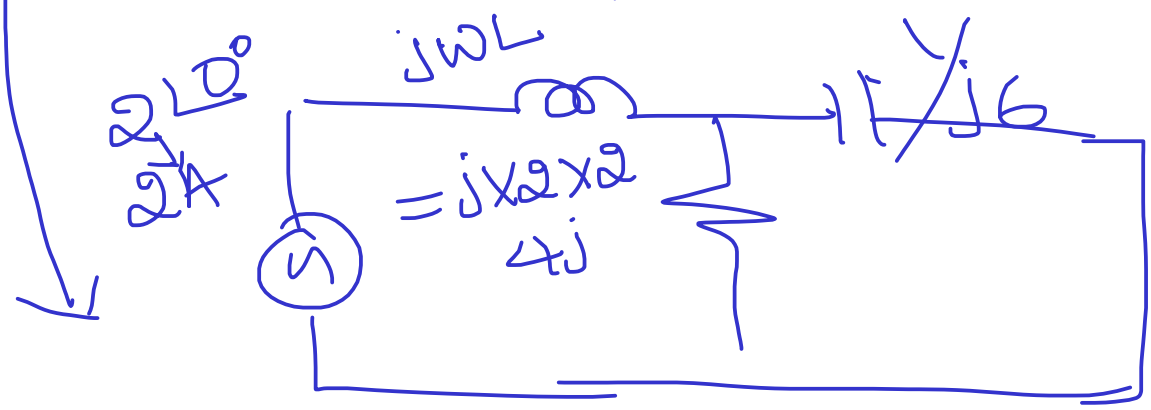
- DC
- Dependent Src
- AC

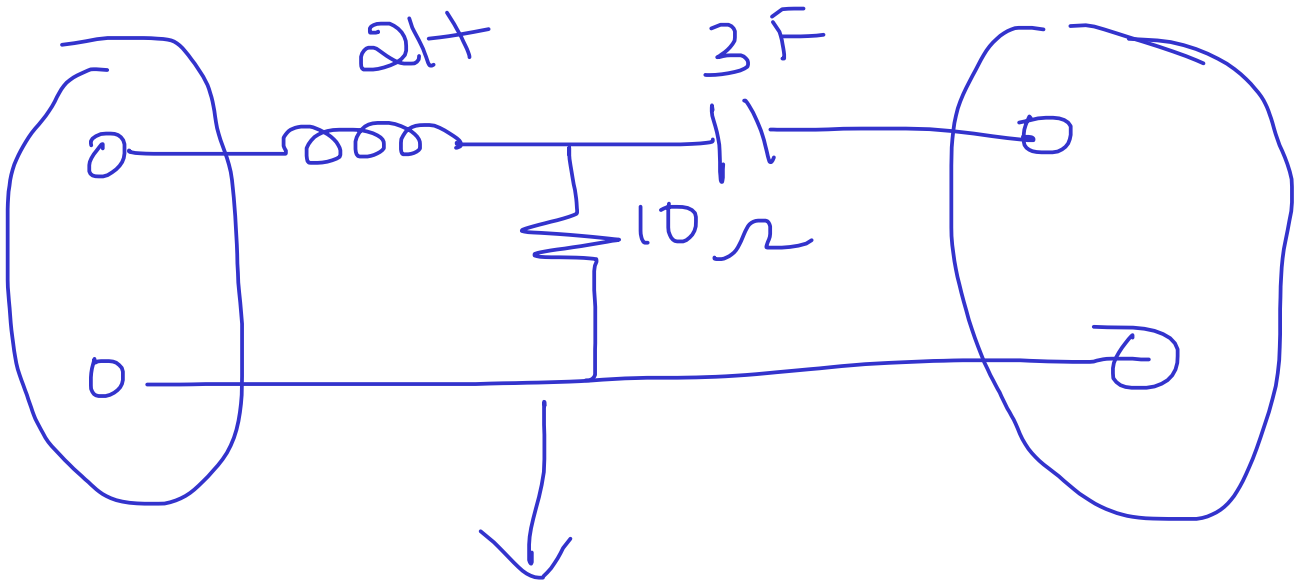


AC

$\frac{1}{j\omega C} = \frac{1}{j \times 2 \times 3}$

f-domain





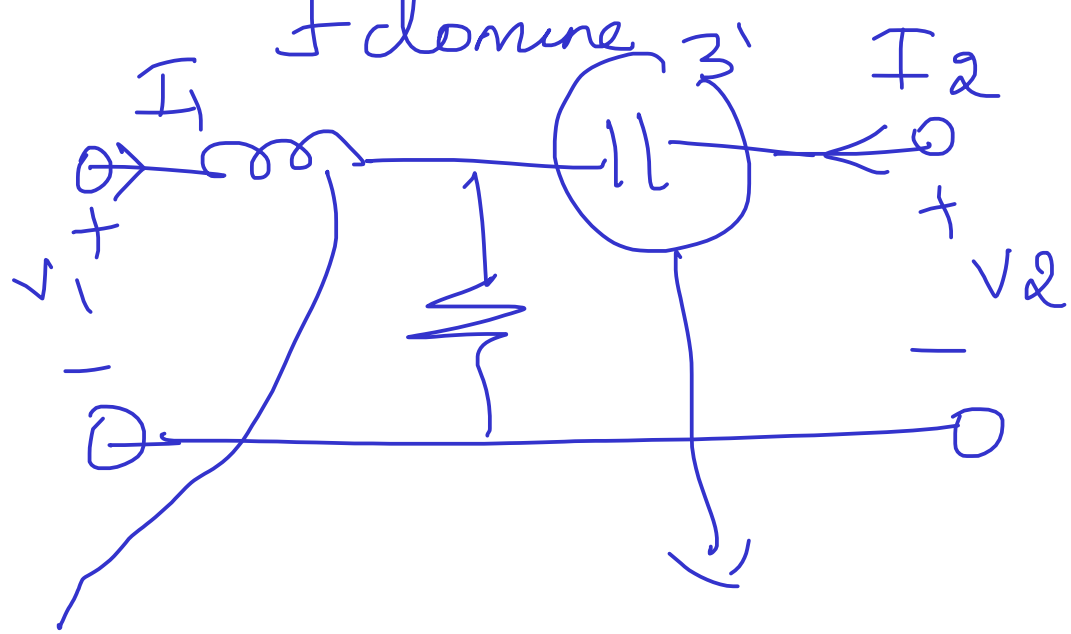
$j\omega L$

$j\omega 2$

$2(j\omega) \rightarrow s$

$2s$

t domine



$3F \rightarrow \frac{1}{j\omega L}$

$\frac{1}{j\omega 3} \rightarrow \frac{1}{3s}$

Cap

freq

$5F \xrightarrow{\text{freq}} \frac{1}{5s}$

$6F \rightarrow \frac{1}{6s}$

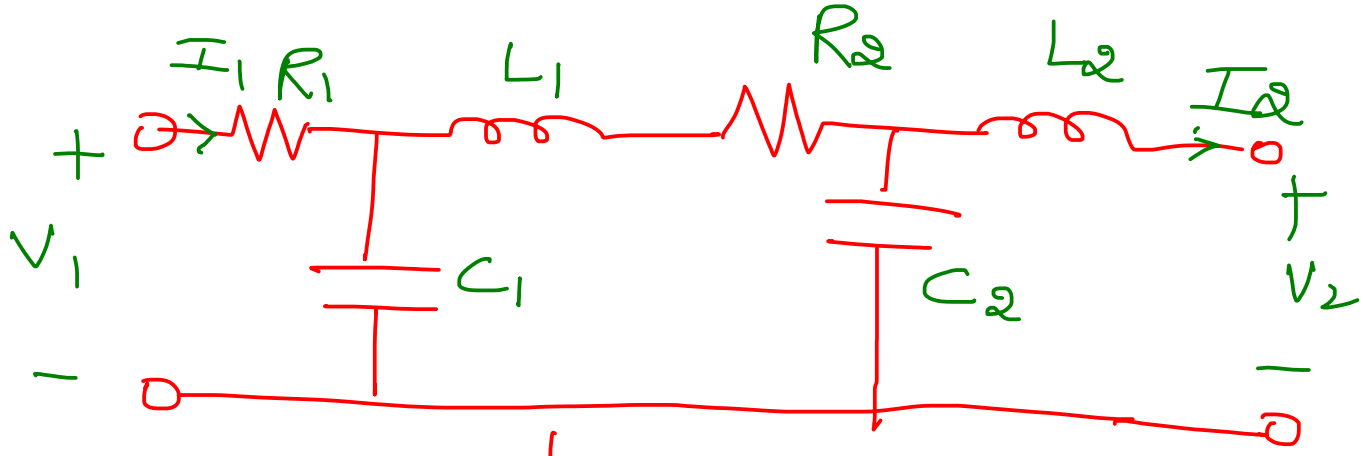
$\frac{1}{6}F \Longrightarrow 6s$

$\cancel{\frac{1}{3}H \rightarrow \frac{1}{3}}$

$3H \rightarrow 3s$,

$\frac{1}{3}H \rightarrow \frac{1}{3s}$

Problem with L & C



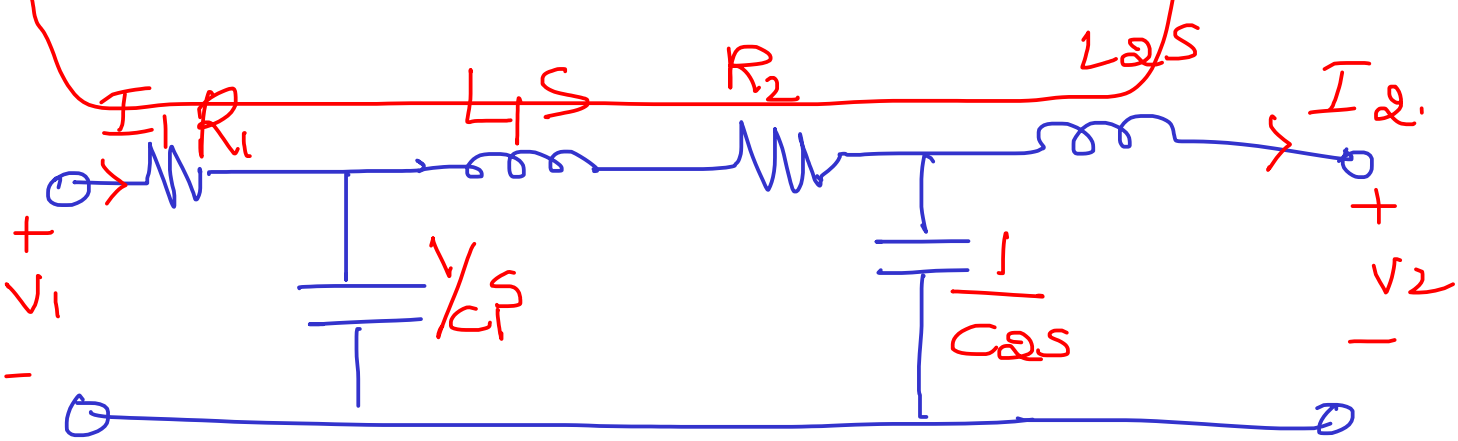
t-domain

$$R \rightarrow R$$

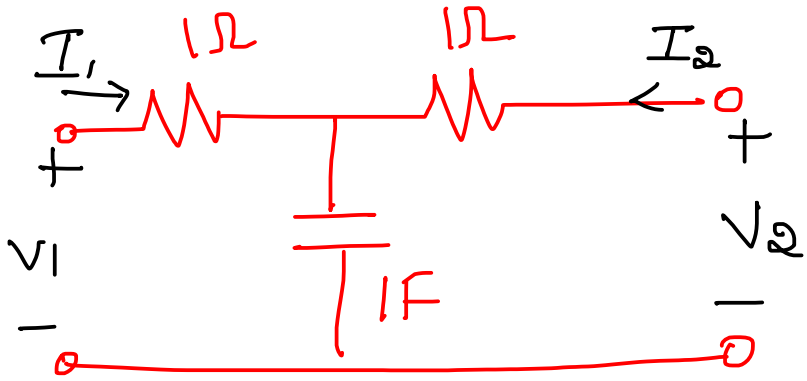
$$L \rightarrow j\omega L \rightarrow sL$$

$$C \rightarrow \frac{1}{j\omega L} \rightarrow \frac{1}{sC}$$

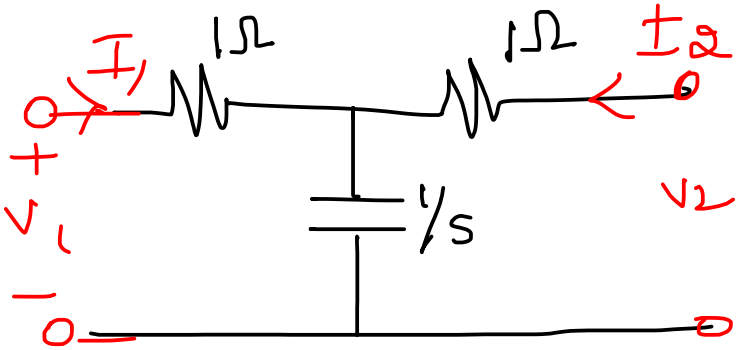
$$\cancel{LT \rightarrow RT \quad s = \sigma + j\omega \rightarrow 0}$$



Ex :- Determine Transmission parameter.



Sol^n:- 2 Port N/w → s-domain

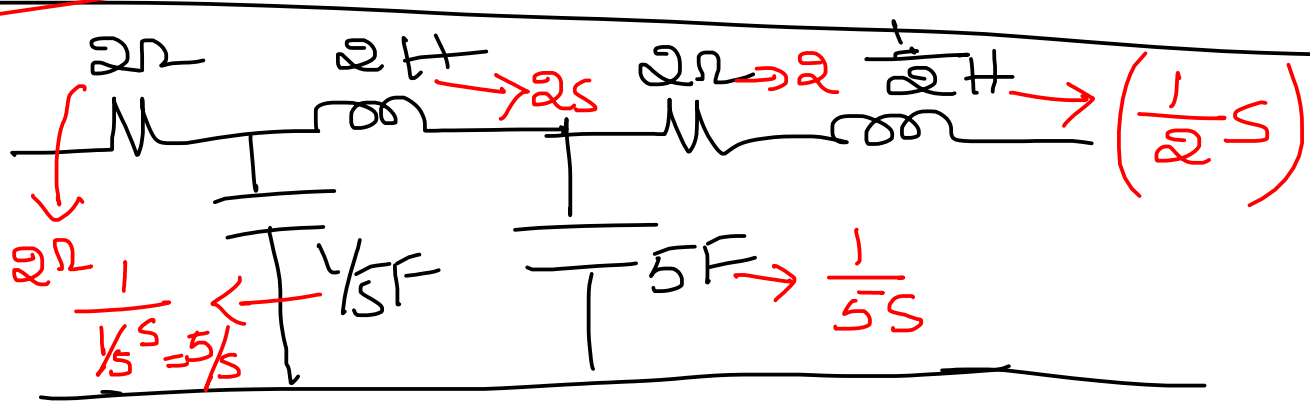


$1\Omega \rightarrow 1\Omega$

$1\Omega \rightarrow 1\Omega$

$1F \rightarrow \frac{1}{s \times 1} = \frac{1}{s}$

other example



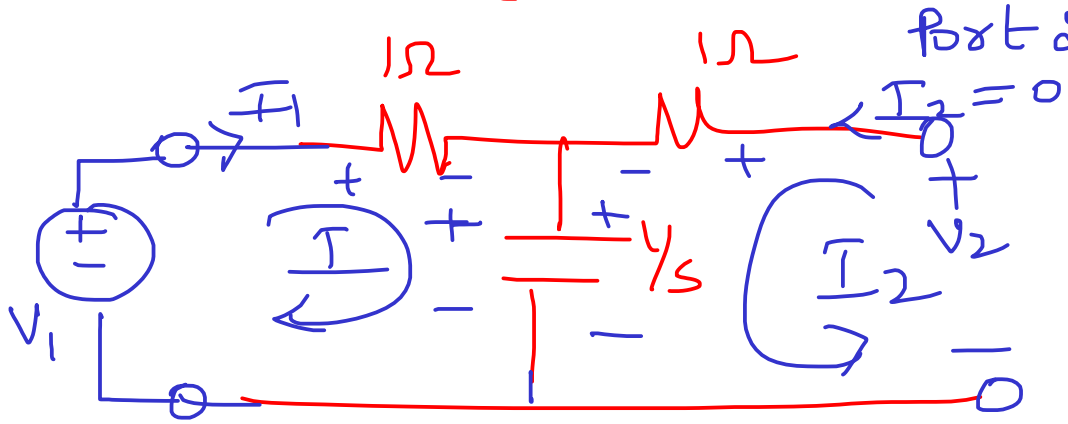
Transmission parameter

$$V_1 = A V_2 - B I_2 \rightarrow (1)$$

$$I_1 = C V_2 - D I_2 \rightarrow (2)$$

Make $\rightarrow I_2 = 0 \longrightarrow$ Port 1 $\rightarrow$ Connect v/g source

Port 2 $\rightarrow$ Open ckt



Apply KVL wrt Mesh 1

$$-V_1 + I_1 \times 1 + (I_1 + I_2)\frac{1}{s} = 0$$

make $I_2 = 0 \Rightarrow V_1 = I_1\left(1 + \frac{1}{s}\right) \longrightarrow (3)$

KVL @ Mesh 2 $\Rightarrow$ $-V_2 + I_2 \times 1 + (I_2 + I_1) \times \frac{1}{s} = 0$

$I_2 = 0 \Rightarrow V_2 = \frac{I_1}{s} \rightarrow (4)$

$\Rightarrow C = \frac{I_1}{V_2} \Rightarrow SV$

To get

$A = \frac{V_1}{V_2} \Rightarrow (3) \Rightarrow I_1 = \frac{V_1}{\left(1 + \frac{1}{s}\right)} \rightarrow (5)$

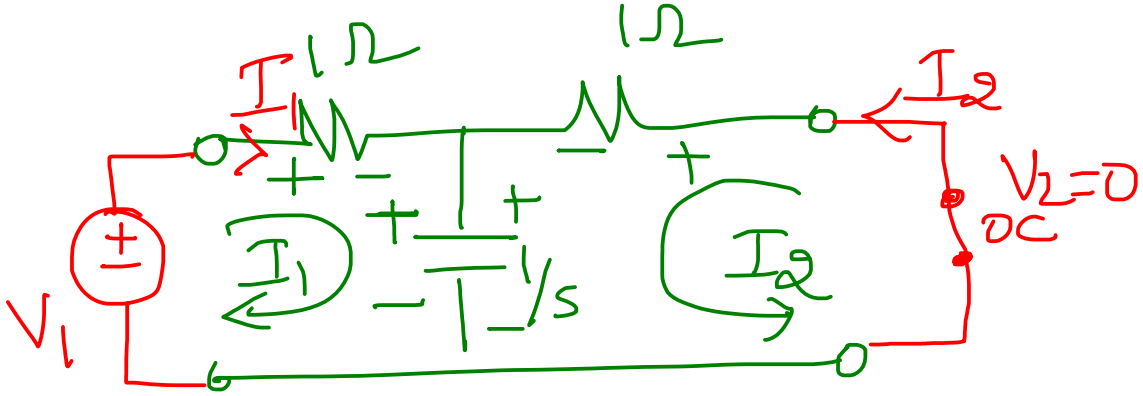
(5)th substituted on (4)

$$V_2 = \left(\frac{V_1}{\left(1+\frac{1}{s}\right)}\right) \times \frac{1}{s}$$

$$\boxed{A = \frac{V_1}{V_2} = s+1}$$

~~Case 2~~:

Make $V_2 = 0 \Longrightarrow$ Port 2 → Short ckt
Port 1



KVL @ Mesh 1

$$-V_1 + I_1(1) + (I_1 + I_2) \times \frac{1}{s} = 0$$

$$V_1 = I_1\left(1+\frac{1}{s}\right) + I_2 \times \frac{1}{s} \quad (4)$$

KVL @ Mesh 2 → $I_2(1) + (I_1 + I_2) \times \frac{1}{s} = 0$

$$D = \frac{I_1}{(-I_2)} = s+1 \quad // \longrightarrow (5)$$

$$\Longrightarrow I_1 = (-I_2)(s+1) \longrightarrow (6)$$

(6) $\xrightarrow{\text{Substituted in eqn}}$ (4)

$$B = \left(\frac{V_1}{-I_2}\right) = (s+2)\,\Omega \quad //$$

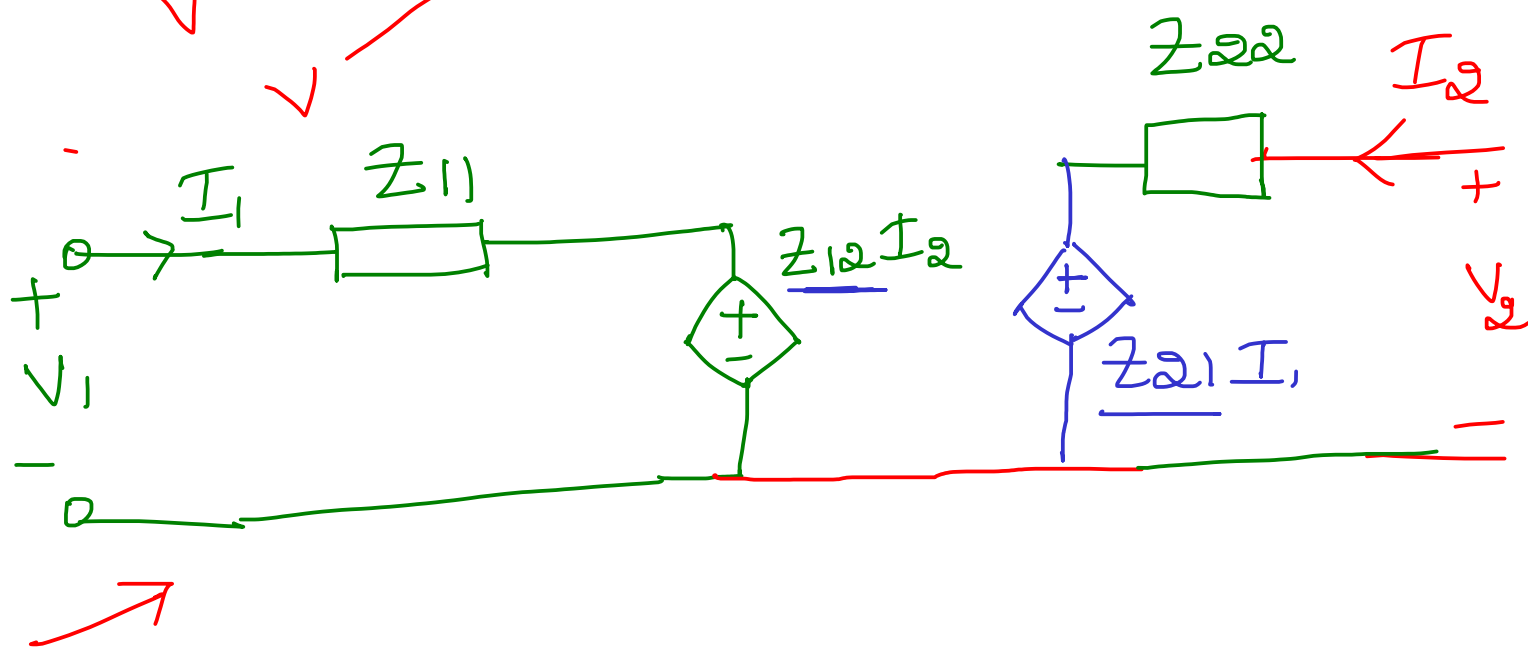
Modelling of Two port parameters

→ Z - parameter

Equational form

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$



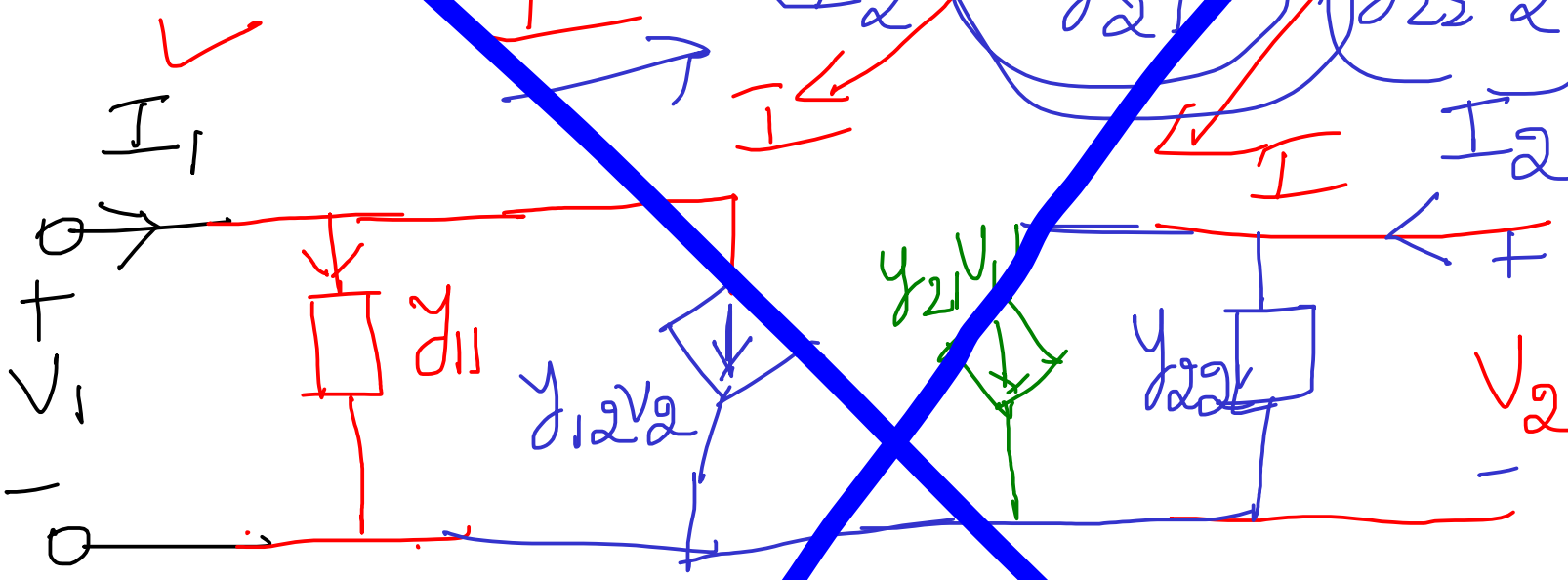
Port 1

Y Parameter

$V = IR \Rightarrow I = \frac{V}{R} = VY$

Equational form $\rightarrow I_1 = y_{11}V_1 + y_{12}V_2$

$I_2 = y_{21}V_1 + y_{22}V_2$

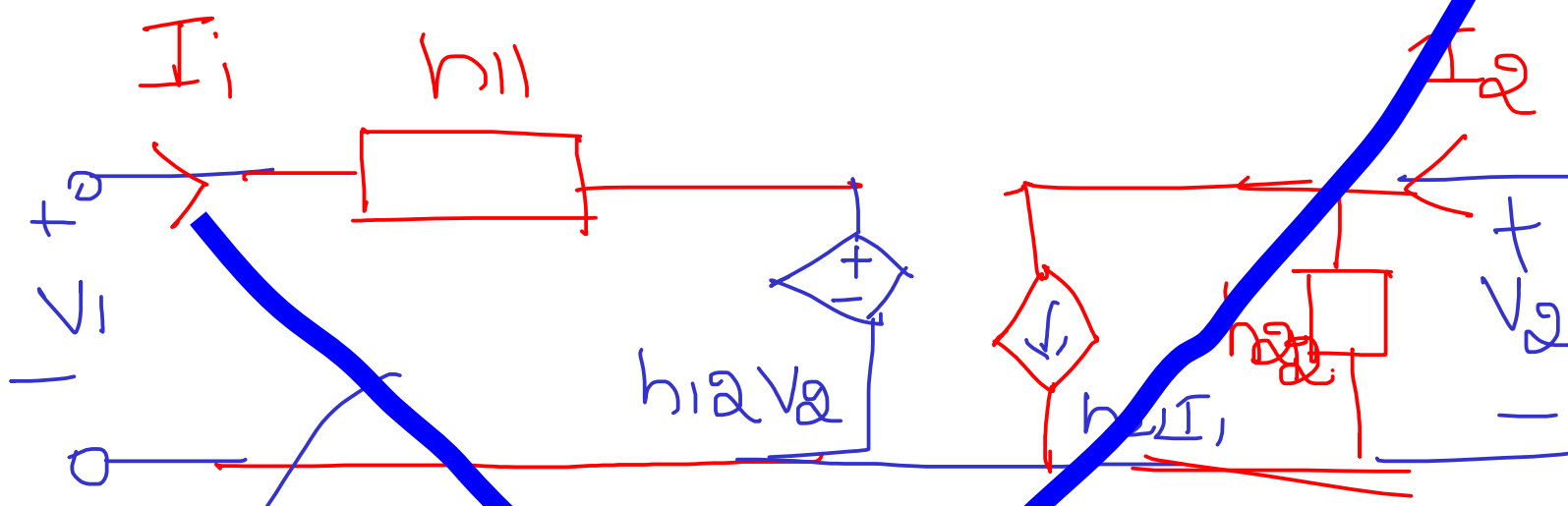


H-Parameter

$V_1 = h_{11} I_1 + h_{12} V_2 \rightarrow (1)$

$I_2 = h_{21} I_1 + h_{22} V_2 \rightarrow (2)$

V V V I I V I



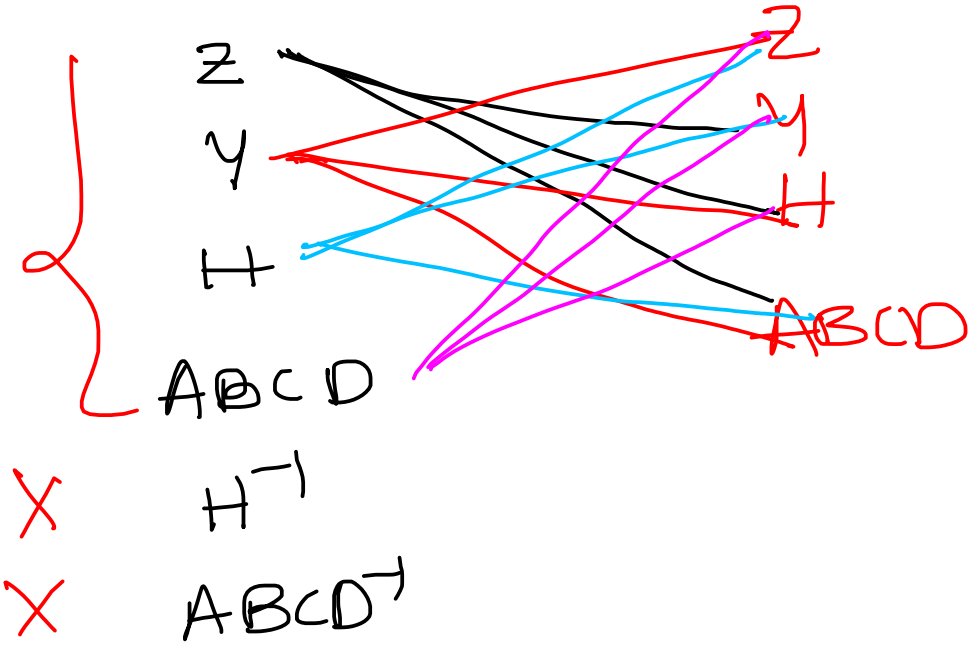
$$h_{11} = \frac{V_1}{I_1} = Z_{11}$$

$$-V_1 + I_1 h_{11} + h_{12} V_2 = 0$$

$$\boxed{V_1 = I_1 h_{11} + h_{12} V_2}$$

<u>NOTE</u>:- No equivalent model for ABCD parameter.

Inter-relation b/w parameters



I) Express Z parameter in terms of all other parameter

(a) Z parameter in terms of Y parameter

Z-parameter $\longrightarrow$ $V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (1)$

$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (2)$

Y-Parameter $\longrightarrow$ $I_1 = y_{11} V_1 + y_{12} V_2 \rightarrow (3)$

$I_2 = y_{21} V_1 + y_{22} V_2 \rightarrow (4)$

Solve ③ & ④ to get v_1 & v_2

↳ Cramer's rule

$$a_1 x + b_1 y = c_1$$
$$a_2 x + b_2 y = c_2$$

To get x & y ⟹ ① find D, D_x, D_y

② $x = \dfrac{D_x}{D}$

$y = \dfrac{D_y}{D}$

$$D = \begin{array}{cc} x & y \\ \hline \end{array} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \qquad D_x = \begin{pmatrix} c_1 & b_1 \\ c_2 & b_2 \end{pmatrix}$$

$$D_y = \begin{pmatrix} a_1 & c_1 \\ a_2 & c_2 \end{pmatrix}$$

$$\rightarrow x = \frac{D_x}{D} \qquad y = \frac{D_y}{D}$$

from eqⁿ (3) & 4

$$D = \begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}$$

$$D_x = \begin{vmatrix} I_1 & y_{12} \\ I_2 & y_{22} \end{vmatrix}$$

$$D_y = \begin{vmatrix} y_{11} & I_1 \\ y_{21} & I_2 \end{vmatrix}$$

$$x = V_1 = \frac{D_x}{D} = \frac{\begin{vmatrix} I_1 & y_{12} \\ I_2 & y_{22} \end{vmatrix}}{\begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}}$$

$$V_1 = \frac{-y_{12} I_2 + y_{22} I_1}{y_{21} y_{11} - y_{12} y_{22}}$$

$$V_1 = \frac{-y_{12}}{y_{21} y_{11} - y_{12} y_{22}} I_2 + \frac{y_{22} \times I_1}{y_{21} y_{11} - y_{12} y_{22}}$$

$$\Delta y$$

$$V_1 = I_1 \times \frac{y_{22}}{\Delta y} - I_2 \frac{y_{12}}{\Delta y}$$

5

Compare Eqn (5) with equation (3)

$$Z_{11} = \frac{y_{22}}{\Delta y} \qquad Z_{12} = \frac{-y_{12}}{\Delta y}$$

$$V_2 = \frac{D_y}{D} = \frac{\begin{vmatrix} y_{11} & \cancel{I_1} \\ y_{21} & I_2 \end{vmatrix}}{\begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}}$$

$$= \frac{I_2 y_{11} - I_1 y_{21}}{y_{11} y_{22} - y_{12} y_{21}}$$

$$\Delta y$$

$$V_2 = +I_1\left(\frac{-y_{21}}{\Delta y}\right) + I_2\left(\frac{y_{11}}{\Delta y}\right)$$

$\longrightarrow$ (6)

Compare (6) with (4)

$$Z_{21} = \frac{-y_{21}}{\Delta y} \qquad Z_{22} = \frac{y_{11}}{\Delta y}$$